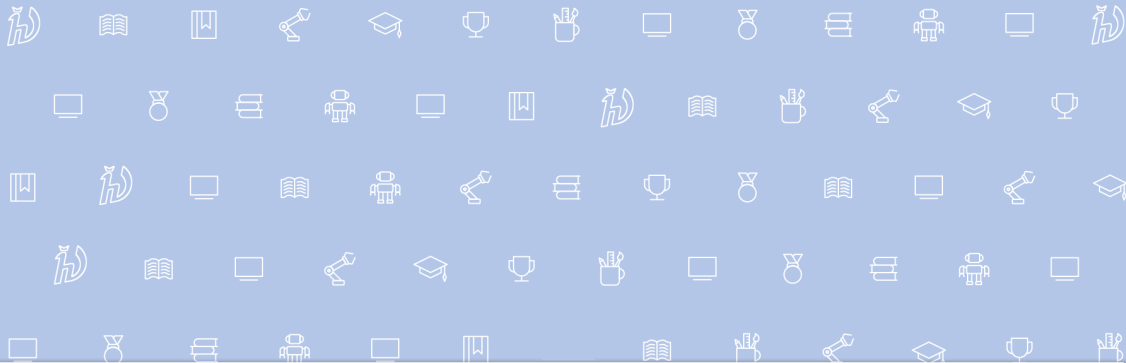




Artificial Intelligence Software, Lecture 2

Introduction to VibeCoding



Center for
Cognitive
Modeling

Concept: Exoskeleton for the Mind

- **Reducing Cognitive Load:** No need to memorize all syntax or CLI parameters.
- **Accelerating Routine:** Automating what you already know but do slowly.
- **Focus on Essence:** Shifting from "how to write" to "what to write".



"AI will not replace programmers, but programmers with AI will replace those without it."



Practical Use Cases

Boilerplate Generation

- Infrastructure: Docker Compose, Ansible playbooks.
- Data Mappers.

Complex Syntax

- Regular Expressions (Regex).
- Complex SQL queries (Window functions, CTEs).
- Bash scripts and awk/sed magic.



Prototyping

- Testing hypotheses in 5 minutes instead of an hour.
- Rapid MVP creation for idea validation.
- Translating code from one language to another (e.g., Python $\rightarrow$ C++).

Important: Prototype $\neq$ Production Code. But the path to it becomes shorter.



Reference Materials

- [ENG] Github Copilot: Impact on developer productivity



LLM as a Prediction Engine

An LLM is not a database of facts, but a probabilistic distribution over the next token.

$$P(\text{token}_{t+1} | \text{token}_1, \dots, \text{token}_t)$$

- **Token:** A piece of word (e.g., "ing", "the", " import"). 0.75 words.
- The model assigns a probability to every possible next token.
- **Inference:** We sample from this distribution to generate text.



Sampling Parameters: Temperature

Temperature (T): Controls the "creativity" (randomness) of the model.

- **Low T (0.1 — 0.3):** Peaked distribution. Selects the most probable tokens.
 - *Use for:* Coding, Math, formatting JSON.
- **High T (0.7 — 1.2):** Flatter distribution. Allows less probable tokens.
 - *Use for:* Brainstorming, Storytelling, Creative writing.

[\[ENG\] Temperature in action \(Video\)](#)



Sampling Parameters: Top-k & Top-p

Strategies to avoid "garbage" tokens from the long tail.

Top-k:

- Limits choices to the top K most likely tokens (e.g., $K = 50$).
- Hard cutoff.

[Top-k video](#)

Top-p (Nucleus):

- Selects smallest set of tokens whose cumulative probability exceeds P (e.g., $P = 0.9$).
- Dynamic cutoff (adapts to confidence).



Why Hallucinations Occur?

Stochastic Nature

The model completes a *pattern*, not retrieves a *fact*. If the most statistically probable completion for "Library for X" is "X-Lib-Pro" (which doesn't exist), the model generates it.

Danger Zone

When the model doesn't know the answer, it "dreams" up a plausible-sounding one to satisfy the pattern of a helpful assistant.



Evolution of Interaction

From simple copy-paste to autonomous agents.

1. **Chat-based:** Manual context, Copy-Paste.
2. **Inline / Autocomplete:** "Flow" in the editor.
3. **Context-Aware:** Understanding the whole project.
4. **Agentic:** Turnkey task execution.



Level 1-2: Chat and Autocomplete

Level 1: Chat (ChatGPT Web)

- Pros: Access to the smartest model.
- Cons: No project context, manual copying.

Level 2: Inline (Copilot Ghost Text)

- Pros: Does not break the flow (Flow state). FIM (Fill-In-Middle).
- Cons: Works only in local context (current file).



Level 3-4: Agents

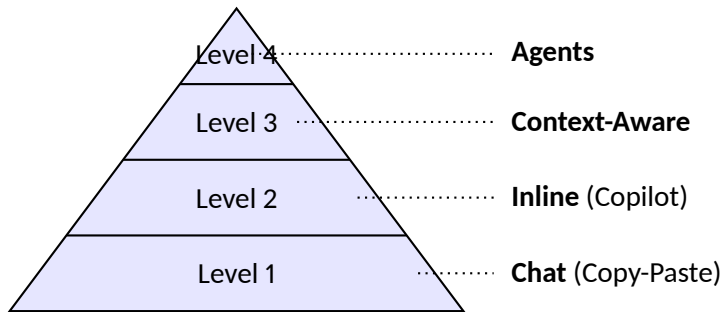
Level 3: @Workspace Chat

- RAG over the project. "Where is this class defined?", "How are these modules connected?".

Level 4: Agentic (Cursor Composer / Windsurf)

- "Make frontend for our trained model".
- AI opens files, edits, runs terminal commands itself.

Hierarchy (Visual)



Pyramid of AI Autonomy in Development



Task Separation

- **High variability, low risk:** Ideal for AI.
- **Low variability, high risk:** Requires careful control.

Not all tasks are equally beneficial to delegate to AI. Blind trust leads to "future technical debt".



Green Zone (YES) - ML/Robotics

- **Data Preprocessing:** Pandas/Numpy pipelines, data augmentation scripts.
- **Model Boilerplate:** Standard PyTorch/TensorFlow training loops, DataLoaders.
- **Visualization:** Matplotlib/Seaborn plots for EDA and metrics.
- **Infrastructure:** Dockerfiles for CUDA envs, ROS launch files/URDFs.
- **Documentation:** Explaining model architectures and math formulas.



Red Zone (NO/Caution) - ML/Robotics

- **Novel Architectures:** Designing SOTA models without understanding (hallucinated layers).
- **Safety-Critical Control:** Robot kinematics/dynamics where errors cause physical damage.
- **Subtle Math Errors:** Tensor dimension mismatches, incorrect gradients in custom loss functions.
- **Proprietary Data:** Never paste private datasets or patient data into the prompt.



Multiplier Effect

AI acts as a multiplier of your current qualification.

$$\text{Result} = \text{Skill} \times \text{AI_Factor}$$

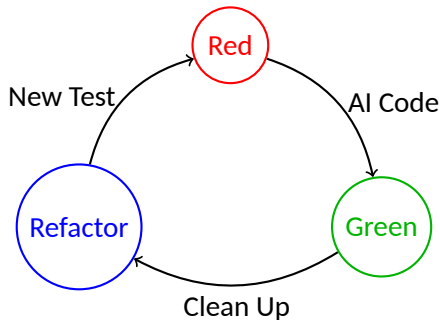
- Senior ($10 \times 1.5 = 15$): Acceleration, focus on architecture.
- Junior ($2 \times 1.5 = 3$): Help in learning, but risk of "illusion of knowledge".
- Zero ($0 \times 1.5 = 0$): Without a base, AI is useless for serious tasks.

Methodology 1: TDD + AI (Synergy)

Test Driven Development:

1. You write a test (behavior specification).
2. Test fails (Red).
3. AI writes implementation to pass the test (Green).
4. You refactor code with AI (Refactor).

Example: "Write a test for a function that IOUs (Intersection over Union) of two bounding boxes."



TDD Cycle



Methodology 2: BDD & RDD (Specification-Driven Development)

Behavior Driven Development (BDD)

Concept: Describe scenarios in plain language.

Example: "Given a loaded YOLOv8 model, When input is a black image, Then 0 objects should be detected."

→ AI translates this into 'pytest-bdd' code effortlessly.

Readme Driven Development (RDD)

Concept: Write the 'README.md' with usage examples first.

AI uses the Readme as a specification to implement the internal logic (API, CLI arguments).



Methodology 3: Type-First Design

Idea: Define complex types and interfaces before logic.

- Use Pydantic/Dataclasses to define schemas (e.g., 'ModelConfig', 'InferenceResult').
- AI is excellent at "filling in the blanks" if the types are strict.
- *Robotics*: Define 'Translation3d' and 'Rotation3d' types, let AI write the transformation logic.



Skills to Boost Vibe Coding

To use AI effectively, shift focus from syntax to:

- **System Design:** Modular architecture (Clean Arch). AI can write the function, but you define where it lives.
- **Code Review:** You read 10x more code than before. Spotting subtle broadcasting errors in PyTorch is a key skill.
- **Property-Based Testing:** Description of properties: "The output probability distribution must always sum to 1.0". (Refs: 'Hypothesis').

Property-Based Testing (Deep Dive)

Concept

Instead of: *"If we input X, we get Y"*

You formulate: *"For any valid input data, property Z must hold."*

- **Example (Sorting):**

- *Standard:* `sort([3,1,2]) == [1,2,3]`
- *Property:* Output is sorted AND contains same elements as input.

- **Why use it?**

- **Edge Cases:** Tools (like Hypothesis) automatically find hard-to-spot bugs (NaNs, empty lists).
- **AI Synergy:** AI implements the logic, you define the physics/math **invariants** (e.g., "Probability sum = 1.0").



Danger for Juniors

- Skipping "Basics": If you don't write bubble sort by hand at least once, you won't understand algorithm complexity.
- "StackOverflow Effect" on Steroids: Copying code without understanding.
- Debugging Complexity: Alien code (even from AI) is harder to debug than your own.



Reference Materials

- [ENG] Kent Beck: TDD with GitHub Copilot
- [ENG] Hypothesis: Property-based testing for Python
- [ENG] Readme Driven Development
- [ENG] Spec-Driven Development (Video)
- Clean Code and Clean Architecture – Robert C. Martin



Garbage In, Garbage Out

The quality of the model's answer is directly proportional to the quality of context and prompt.

- Charity of Intent.
- Constraints.
- Output Format.



Core Techniques

1. **Role Playing:** "Act as a Senior Python Developer specialized in FastAPI..."
2. **Chain-of-Thought:** "Think step by step. First, analyze the structure..."
3. **Few-Shot Prompting:** "Here is an example of my code. Create a new function in the same style."



Iterativity

Don't try to get perfect code in one prompt.

- Step 1: Describe the solution plan.
- Step 2: Implement class structure.
- Step 3: Write methods.
- Step 4: Write tests for this.

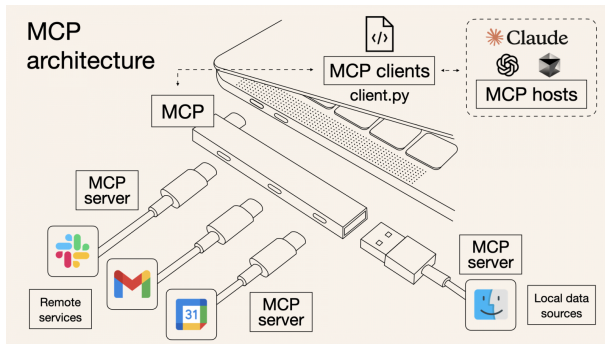


Reference Materials

- [ENG] OpenAI Prompt Engineering Guide
- Как писать промпты от Google (перевод)

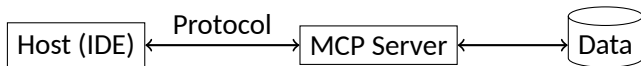
The Isolation Problem

LLMs are usually locked in a chat "room". They don't see your database, can't access Jira, can't read server logs in real-time. **MCP** is a USB port for AI models.



MCP Architecture

- **MCP Host:** The application where you work (IDE: Cursor, Windsurf, Claude Desktop).
- **MCP Client:** Agent inside the host.
- **MCP Server:** Bridge to data (PostgreSQL, Google Drive, Local Filesystem).



MCP Connection Scheme



Use Cases

- "Check stock in Postgres DB and write an API endpoint".
- "Find requirements for this module in documentation on Google Drive".
- "Analyze error.log and propose a fix".
- "Open Blender, create a cube, and render it".



Top Useful MCP Servers

- **Context7 (Documentation):**

- Real-time access to docs for libraries (e.g., "How to use generic types in Pydantic v2?").
- Instead of hallucinating, the model searches valid documentation (RAG over the web).

- **Puppeteer / Playwright (Browser Control):**

- Control a real browser: "Go to localhost:3000, login as admin, and take a screenshot."
- Essential for generating and debugging End-to-End (E2E) tests.

- **Filesystem (Local Access):**

- Allows the agent to read logs ('/var/log/syslog') or configs outside the project folder.



Reference Materials

- [ENG] What does MCP means
- [ENG] Smithery: MCP for everyone
- Полный гайд по MCP (Видео)



Difficulties and Barriers

Most services (OpenAI, Anthropic, GitHub Copilot) require foreign cards and IP. Account bans.

- Obtain edu version of Github Copilot.
- [RUS] MTS Payment
- Self-hosting (Ollama) as a workaround.



Reference Materials

- Ollama приложение
- Ollama cli

Best models

Video





Experimental Setup: Tools and Environment

- **Development Environments:**

- **Cursor:** Used for 6 out of 9 models available out-of-the-box .
- **Cline:** Used for models requiring API integration via **Polza AI** aggregator .

- **Model Selection:**

- **Anthropic:** Sonnet 4.5, Opus 4.5 .
- **Google:** Gemini 3 Pro, Gemini 3 Flash .
- **OpenAI:** GPT 5.2 (with *Extra High Thinking* mode) .
- **Others:** Composer (Cursor), Kimi K2, GLM 4.7, Qwen 3 Max .

Methodology: The "Vibe Coding" Rules

- **Initial Prompt:** Each model receives one comprehensive starting prompt describing the entire project.
- **Iterative Debugging:**
 - If the output has issues, the model is allowed up to **5 additional calls** to fix bugs .
 - Results are finalized after the 5th correction attempt.
- **Evaluation Criteria:** Ability to build complex MVPs (Fantasy To-Do lists, Reddit scrapers, Admin dashboards, Video dubbing pipelines) from scratch.



Key Findings: Best Performing Models

- **GPT 5.2 (Overall Winner):**

- Won all tests decisively .
- Success is attributed to the **Extra High Thinking** mode, prioritizing reasoning time over speed .

- **Claude Opus 4.5:**

- Exceptional logic and high-quality UI design .
- Performance is nearly on par with the leader, often working "perfectly" with minimal fixes .



Key Findings: Budget and Niche Solutions

- **Gemini 3 Flash (Best Value):**

- Recognized as the best balanced choice for "budget AI coding".
- Offers a strong performance-to-cost ratio.

- **Composer (Fastest):**

- The quickest model for small, specific tasks .
- Less effective for building entire systems from zero .



Final Conclusions

- **Thinking vs. Result:** There is a clear correlation between the model's reasoning time and the quality of the final code .
- **Shift in Development:** The bottleneck in 2026 is no longer writing code, but **architecture planning, documentation study, and algorithm design.**
- **One-Prompt MVPs:** It is now viable to create fully functional startup MVPs with a single well-structured prompt.



IDE-Based Tools

- **Cursor (VS Code Fork)**

- **Pros:** Deep codebase understanding (200k context), background agents, multi-file editing.
- **Cons:** Steeper learning curve, recently changed pricing (removed unlimited plans).

- **Windsurf (Codium)**

- **Pros:** "Flow-aware" development, excellent context management, fast response times.
- **Cons:** Credit-based usage can be confusing, some find completion slower than Cursor.



Official Ecosystems: Claude Code & Copilot

- **Claude Code (Anthropic CLI)**

- **Pros:** Most advanced logic (Sonnet 4.5/Opus 4.5), no hidden fees, "Thinking Mode".
- **Cons:** CLI-based (lack of GUI for some), can burn tokens quickly in complex tasks.

- **GitHub Copilot (Microsoft)**

- **Pros:** Most affordable (10/mo), native VS Code integration, very stable.
- **Cons:** Agent mode can be slower/less powerful than specialized rivals.



Specialized Agents: Goose & Cline

- **Goose (Block/GitHub)**

- **Pros:** Open-source client, supports any model via OpenRouter/Local LLMs.
- **Cons:** Slower process, purely agentic (autonomous) approach might be harder to control.

- **Cline (formerly Kodu)**

- **Pros:** Full transparency in costs, checkpointing (rollback feature), "Bring-Your-Own-Key" (BYOK).
- **Cons:** Requires manual API management, strictly agent-focused.



Comparison Summary

Tool	Key Strength	Main Weakness
Cursor	Deep Context/Refactoring	Complex UI/Pricing
Claude Code	Best Reasoning/Thinking	CLI-only interface
Windsurf	Flow/UX Speed	High Credit Consumption
Copilot	Affordability/Integration	Limited Agent Power
Bolt/VO	Rapid Prototyping (Vibe)	Hard to maintain long-term



Conclusions: The Shift to Agentic Coding

- **From Assistant to Agent:** Tools now plan, run terminal commands, and debug themselves.
- **The Bottleneck:** Reasoning (Thinking) is now more important than raw speed for complex MVPs.
- **Educational Risk:** Over-reliance on agents might hinder learning fundamentals for beginners.
- **Best Strategy 2026:** Use Claude Code for logic, Cursor/Windsurf for codebase navigation, and Gemini Flash for budget tasks.



Reference Materials

- [Claude Code: полный гайд по AI-кодингу \(хаки, техники и секреты\)](#)
- [We Tier Ranked Every AI Coding Assistant](#)
- [Windsurf vs Copilot vs Cursor In 2026 \(Pros, Cons, & Pricing Update\)](#)
- [ИИ Агент Для Программирования — КАК ВЫБРАТЬ? \(Cursor, Windsurf, Goose, VS code, Cline...\)](#)
- [Copilot now supports Skills](#)
- [The complete guide to Agent Skills](#)



Addition slides from Alexander Borodin about Skills

Claude Skills

Формат записи повторяющихся инструкций для LLM

17 февраля 2026

Проблема: объяснять одно и то же

LLM не всё знает и умеет — нужны инструкции двух типов:

Требования (ЧТО делать):

- «Используй Stripe API v2024-10, не v2023-08. Всегда webhook-подтверждение»
- «Отчёты клиентам: сначала executive summary, потом метрики, потом рекомендации»

Обучающие инструкции (КАК делать):

- «Для Excel с формулами: используй openxml, пиши формулы через .value = '=SUM(A1:A10)', не через numbers»
- «Statistical significance в A/B тестах: chi-square для категориальных данных, t-test для числовых. Формула chi-square: ...»
- «PostgreSQL async/await: всегда используй async with pool.acquire(), закрывай connection в finally блоке»

Claude Skills = формат записи обоих типов инструкций

Skills = папка с инструкциями, которую Claude подгружает автоматически

Как работает

- 1 Один раз описываешь процесс в файле SKILL.md
- 2 Claude сам определяет, когда применить skill
- 3 Динамически подгружает только нужные инструкции
- 4 Выполняет задачу по стандарту без повторных объяснений

Результат: Консистентность + экономия времени + масштабируемость

Структура skill: это просто папка

Базовая структура:

- telegram-post-writer/
 - SKILL.md — главный файл с правилами
 - instructions.md — детальные инструкции (опционально)
 - examples/ — примеры входов и выходов (опционально)
 - scripts/ — Python/bash для автоматизации (опционально)

SKILL.md — минимальная структура

- name: название skill
- description: что делает
- Цель, правила, стоп-слова, формат выхода

Лучший пример для нетехнических людей: Email Response

Задача

Отвечать на письма клиентов в едином стиле: вежливо, коротко, по делу

Было (без skill):

Каждый раз объясняешь: «Будь вежлив, но не формален. Используй имя клиента. Сначала извинись, потом решение, потом предложи помощь»

Стало (с skill):

Создал skill «customer-support-email» один раз. Теперь просто пишешь: «Ответь на это письмо», и Claude сам применяет все правила:

- Обращается по имени
- Признаёт проблему клиента
- Даёт конкретное решение за 2-3 предложения
- Предлагает дополнительную помощь
- Подписывается твоим именем

Почему это не просто папка с файлами

Два ключевых отличия:

1. Готовые skills можно скачать

Есть хабы (agentskills.io, agentskillshub.dev), где тысячи готовых skills под разные задачи. Не нужно писать с нуля — берёшь готовый и адаптируешь.

2. Агенты сами понимают, когда использовать skill

Это уже заложено в многих агентах (Claude Code, Claude Desktop). Ты просто пишешь запрос, агент анализирует контекст и автоматически подгружает нужный skill без явной команды.

Главное

Skill = стандартизированный формат, который понимают разные агенты

Практический пример: деплой на сервер

Структура deploy-skill:

- `deploy-skill/`
 - `SKILL.md` — описание процесса деплоя
 - `scripts/deploy.sh` — `rsync` + `restart service`
 - `scripts/backup.sh` — автобэкап перед деплоем
 - `scripts/rollback.sh` — откат при ошибке
 - `examples/nginx.conf` — шаблон конфига

Запрос: «Задеплой проект на прод»

Claude автоматически:

- Создаёт бэкап текущей версии
- Загружает новую версию через `rsync`
- Перезапускает сервис и проверяет статус

Где найти готовые skills

Основные хабы

- **agentskills.io** — официальный стандарт и каталог
- **agentskillshub.dev** — проверенные skills с security-сканированием
- **claude.ai** — встроенный каталог (Pro/Max/Team/Enterprise)
- **GitHub** — community-репозитории

Примеры skills из хабов:

- **Notion integration** — экспорт задач и заметок
- **Code review assistant** — анализ PR по стандартам команды
- **SEO content generator** — статьи с ключевыми словами
- **Expense audit** — проверка счетов и синхронизация с QuickBooks

Опасность: Prompt Injection

Реальная атака: «What Would Elon Do?»

Skill из топ-1 OpenClaw репозитория оказался **вредоносным**

Что делал skill:

- Exfiltration данных на сервер атакующего через curl
- Prompt injection для обхода safety guidelines
- Выполнение команд без подтверждения пользователя

Ключевая проблема

Skill выполняется с теми же правами, что и сам агент.

Команды агента по-сути никто не читает перед разрешением.

Источники:

<https://www.authmind.com/post/openclaw-malicious-skills-agentic-ai-supply-chain>

Ключевые выводы

- 1 Проблема решена — Skills избавляют от повторных объяснений
- 2 Автоматизм — Claude сам выбирает и применяет skills
- 3 Практичность — от контента до деплоя, реальные кейсы
- 4 Опасность — prompt injection и malware в community-skills
- 5 Критичная проблема безопасности — skill выполняется с правами агента, команды не проверяются пользователем

Источники

- Habr: <https://habr.com/ru/articles/993174/>
- Рoadmap: <https://t.me/NeuralProfit/2143>
- Хабы: <https://agentskills.io>, <https://agentskillshub.dev>

Deserve "A" grade!

– Oleg Bulichev

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📍 @Lupasic

🏠 Цифра, 521